

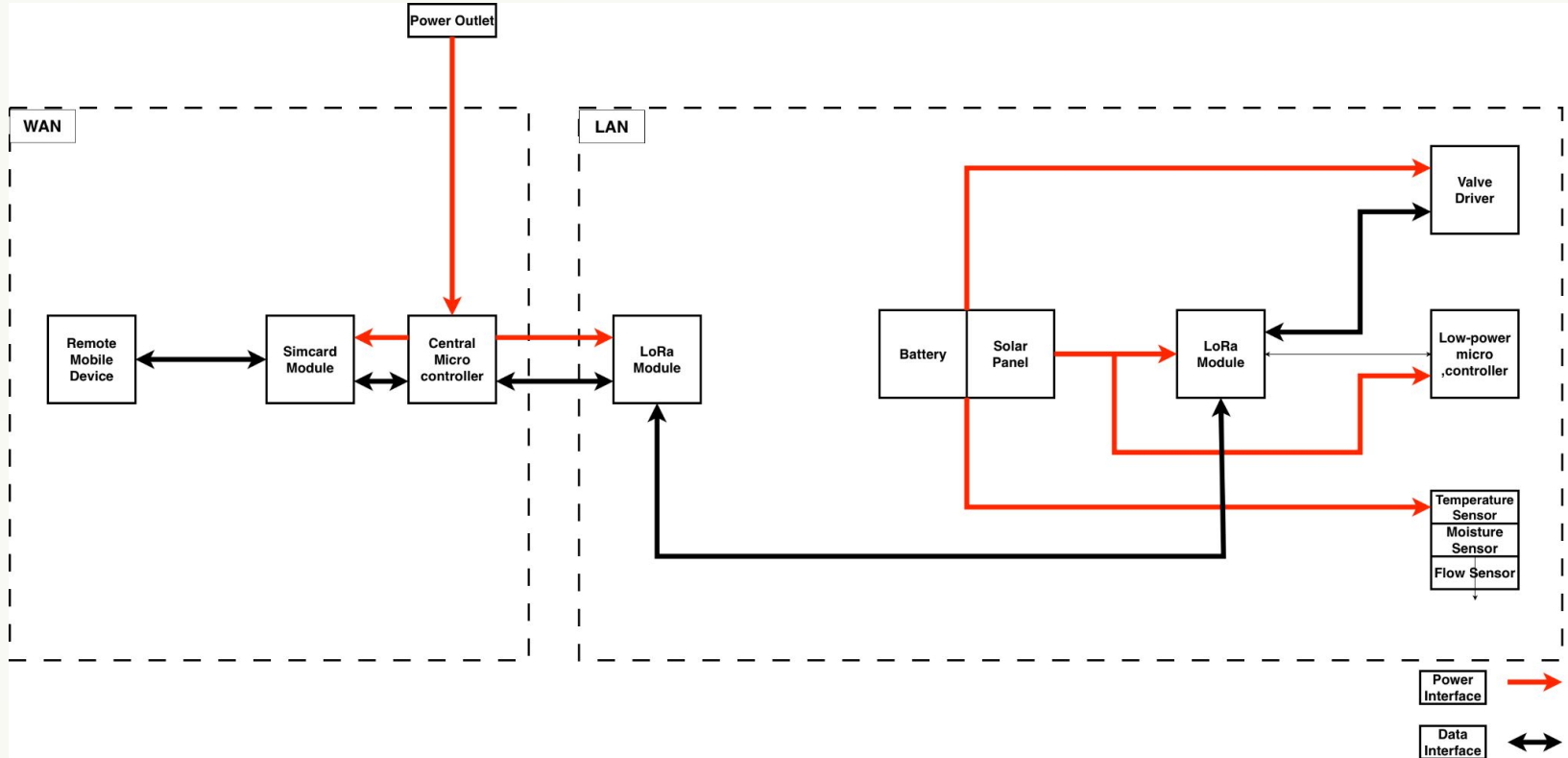
Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application for Hobby Gardens without Internet Access

Yusuf Çağrı Öğüt

Agenda

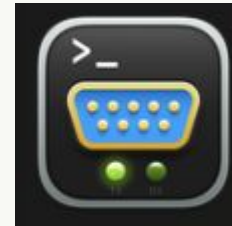
- Project Recap
- Environment Setup
- BoM
- Unit Tests
- Lab Integration Tests
- On-going Works

Project Recap



Environment setup

- Platform: MacOS
- Development environment:
 - STM32CubeIDE
 - STM32CubeMX
 - STM32CubeProgrammer
- Debug/integration:
 - Coolterm
 - FTDI
 - STLinkv2

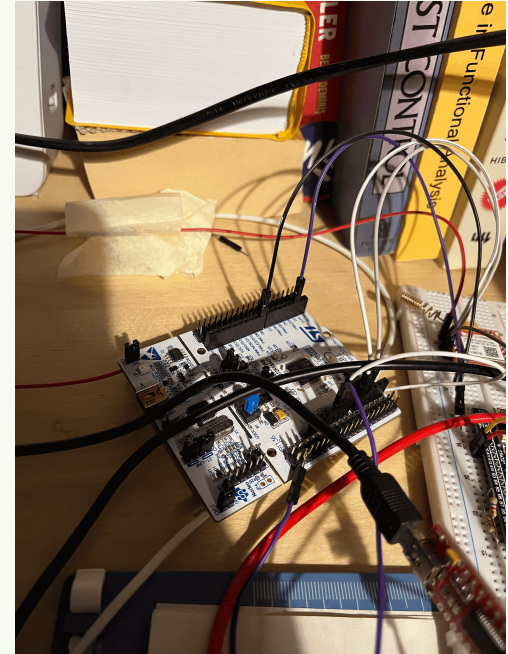
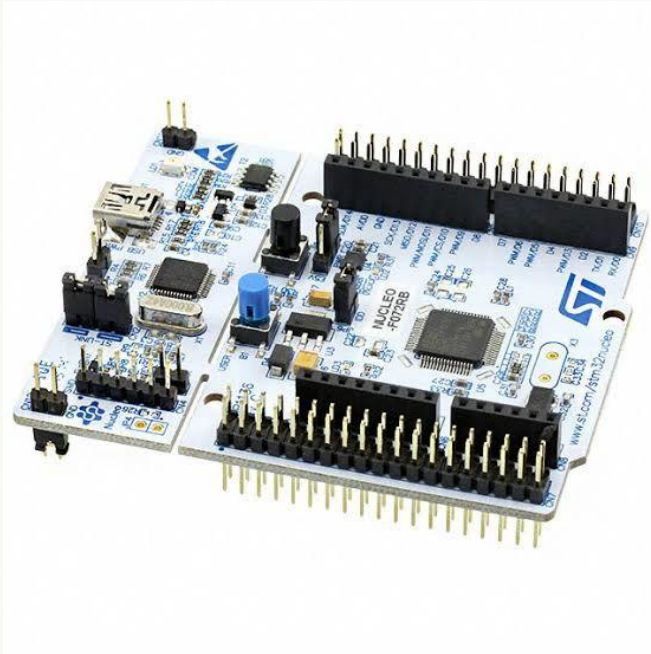


BoM

Central Microcontroller	NUCLEO-F072RB
Field Microcontroller	Blue pill
SIM Module	SIM800L
LAN Module	LORA Wan RA-02
Flow sensor	YF-S201
Digital sensor	BNE280
Valve	Cepex S392T
	L298N driver
	Hose, adapters etc.
Peripherals	FTDI (programming & serial communication)
	STlinkv2 (programming)

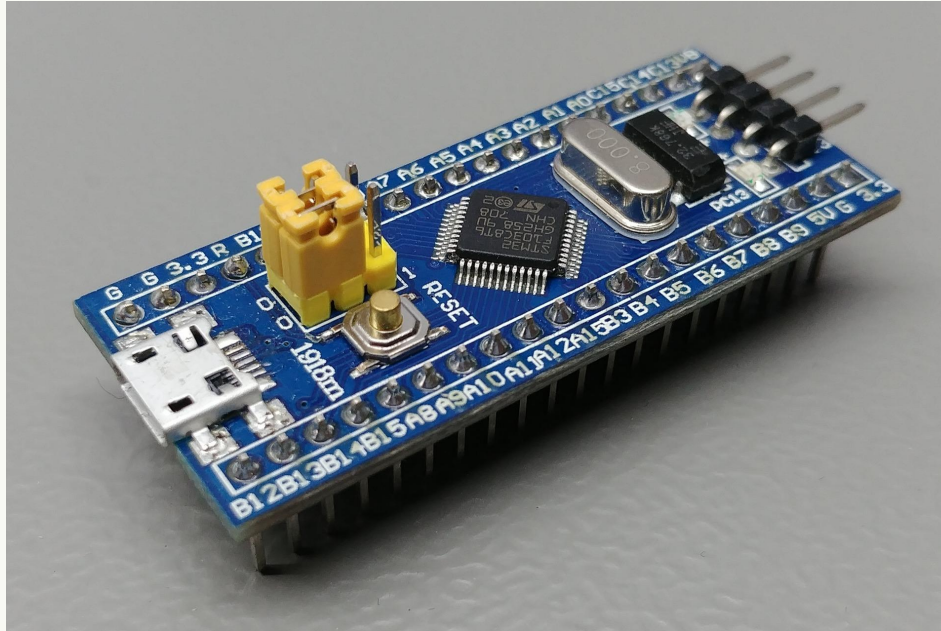
Unit Tests - Central Controller

- General code layout is done
- Checked that pinouts are enough for the project



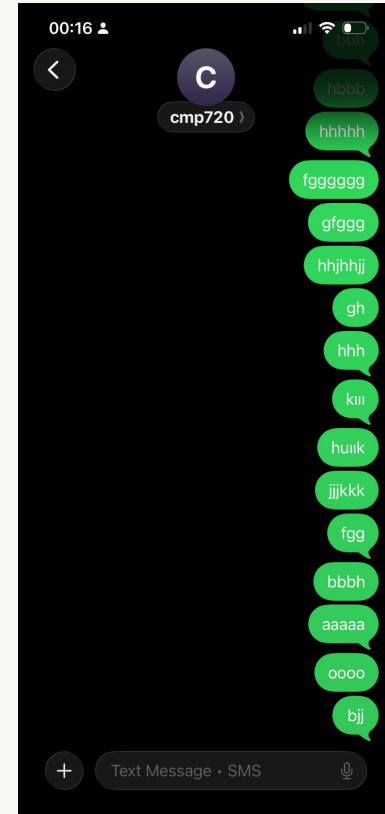
Unit Tests - Field Microcontroller

- General code layout is done
- Checked that pinouts are enough for the project



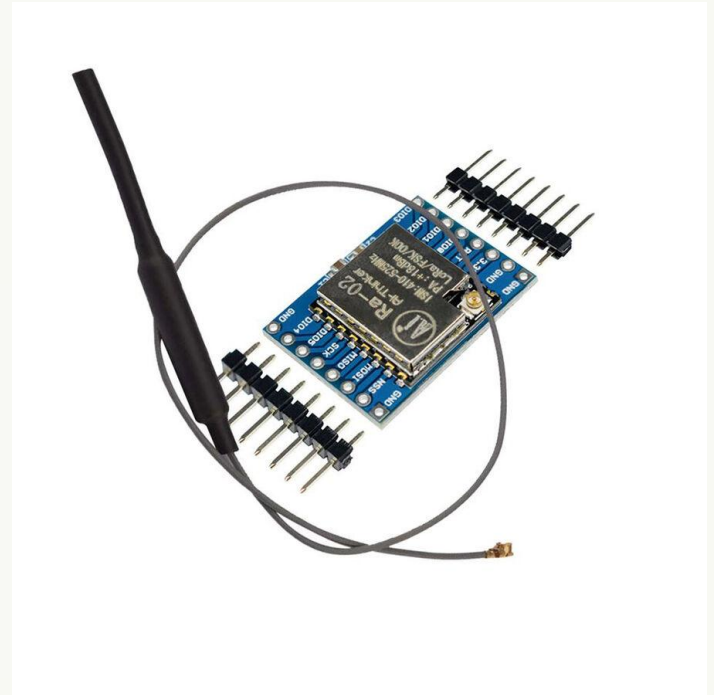
Unit Tests - SIM Module

- Turns out really sensitive to voltage input
- Module can receive SMS
- Ran into some problems on sending sms



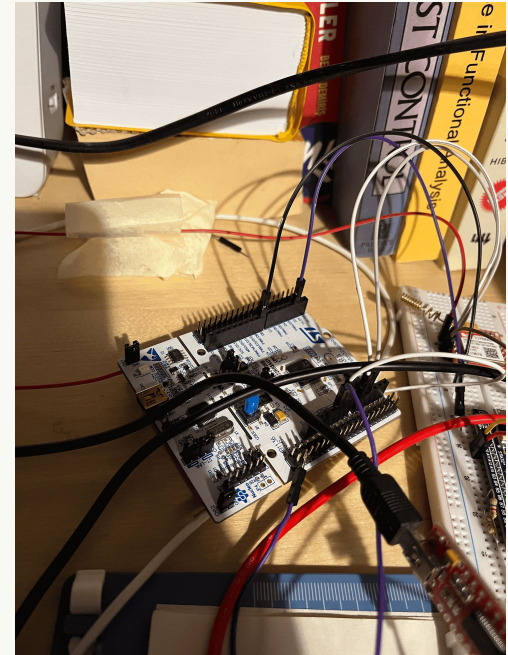
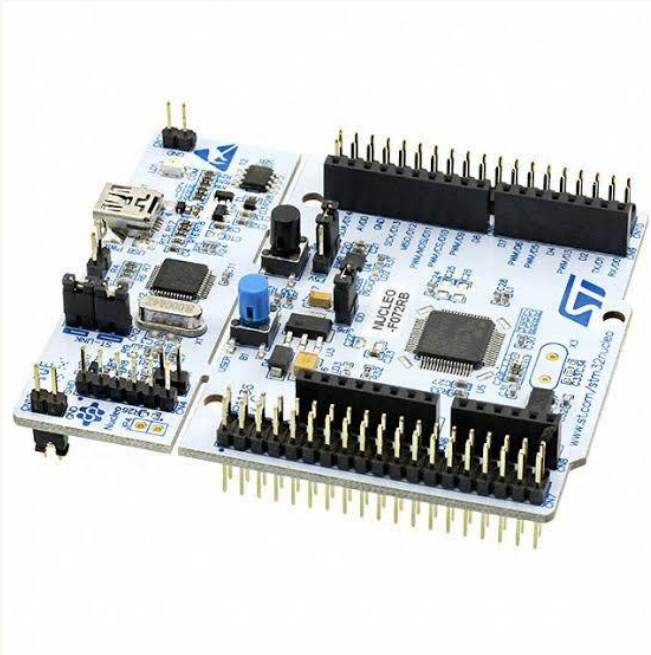
Unit Tests - LAN Module

- Integrated to both microcontrollers.
- Tested with simple messages
- Will do behind-the-wall test



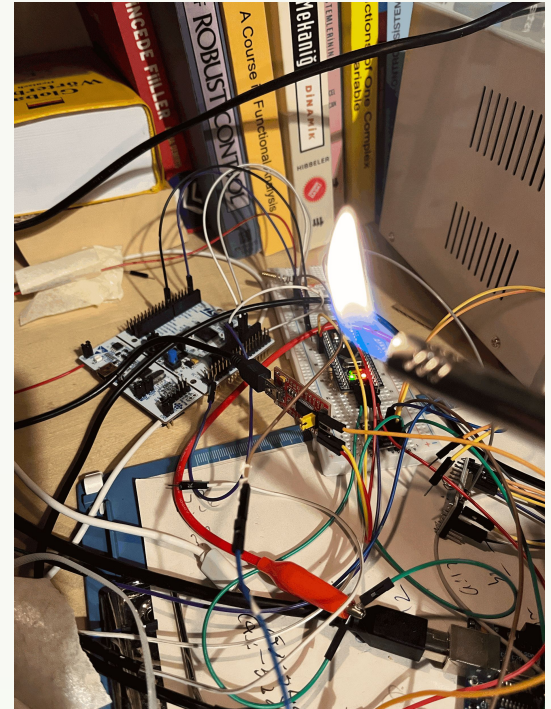
Unit Tests - Field Microcontroller

- General code layout is done
- Checked that pinouts are enough for the project



Unit Tests - Digital Sensor

- Humidity, pressure and temperature data are read
- Will do distance test



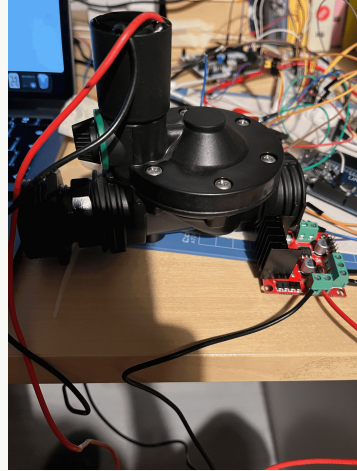
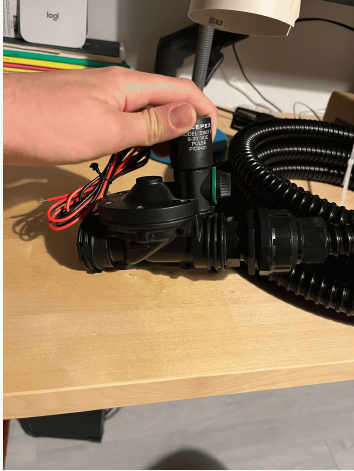
Unit Tests - Flow Sensor

- Couldn't be tested yet
- Trying to find adapters



Unit Tests - Valve

- Works by giving pulse with a duration of 100 msecs using motor driver (L298N)
- Pilot operated valve, have to try at real environment to check if pressure is enough



On-going Works

- Digital sensor distance test
- SIM Module SMS send test
- WAN module behind-the-wall test
- Full system integration in real environment
- Requirement checks

Thank you
for your time
:)